Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

An achromatic doublet consists of two lenses of focal lengths f_1 and f_2 whose materials have dispersive powers in the ratio $2/3$. If the focal length of the doublet is 54 cm, the values of f_1 and f_2 respectively are

(A) 27 cm, -18 cm

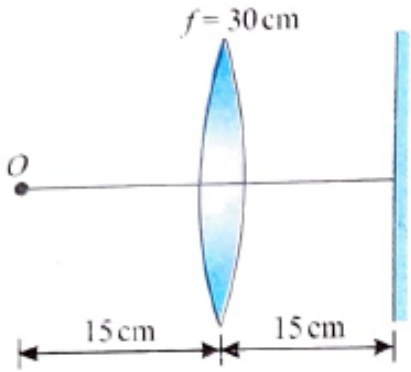
(B) 16 cm, -30 cm

(C) 18 cm, -27 cm

(D) -15 cm, 20 cm

Q.2

An object O is kept in front of a converging lens of focal length 30 cm behind which there is a plane mirror at 15 cm from the lens. Choose the correct statements.



(A) The final image is formed at 60 cm from the lens towards right of it.

(B) The final image is at 60 cm from lens towards left of it.

(C) The final image is real.

(D) Both 2 and 3 are correct.

Q.3

A person can see objects clearly only upto a maximum distance of 50 cm. His eye defect, nature of the corrective lens and its focal length are respectively

(A) Hypermetropia, concave, 50 cm

(B) Myopia, convex, 50 cm

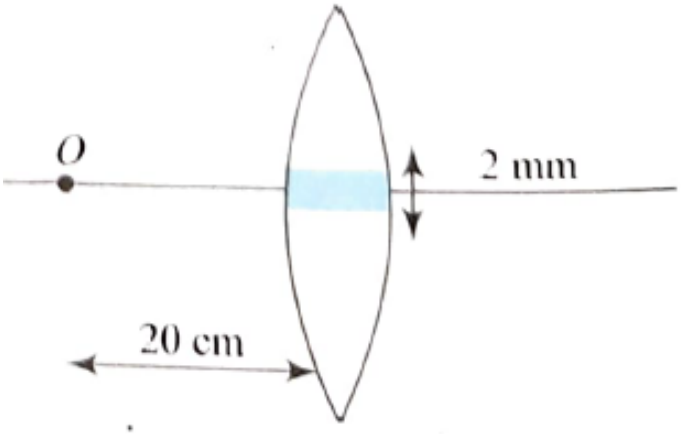
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(C) Myopia, concave, 50 cm

(D) Catract, convex, 50 cm

Q.4

A convex lens of focal length $10\ \text{cm}$ is painted black at middle portion as shown in figure. An object is placed ar distance of $20\ \text{cm}$ from the lens.



Then,

(A) only one image will be formed by the lens

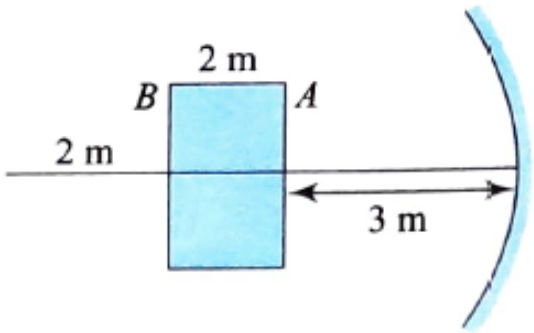
(B) the distance between the two images formed by such a lens is $6\ \text{mm}$

(C) the distance between the images is $4\ \text{mm}$

(D) the distance between the images is $2\ \text{mm}$

Q.5

A cube of side $2\ \text{m}$ is placed in front of a concave mirror of focal length $1\ \text{m}$ with its face A at a distance of $3\ \text{m}$ and face B at a distance of $5\ \text{m}$ form the mirror. The distance between the images of faces A and B and heights of images of A and B are, respectively,



(A) $1\ \text{m}, 0.5\ \text{m}, 0.25\ \text{m}$

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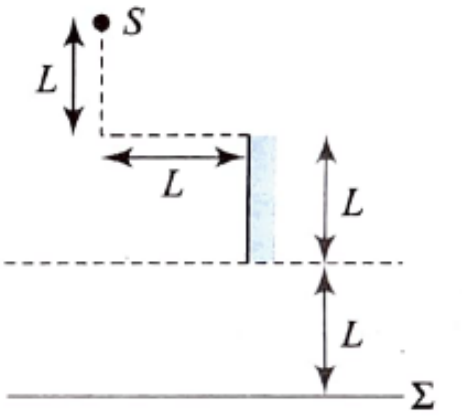
(B) $0.5\ \text{m}, 1\ \text{m}, 0.25\ \text{m}$

(C) $0.5\ \text{m}, 0.25\ \text{m}, 1\ \text{m}$

(D) $0.25\ \text{m}, 1\ \text{m}, 0.5\ \text{m}$

Q.6

A point source of light is placed in front of a plane mirror as shown in the figure.



Determine the length of reflected path of light on its screen Σ.

(A) L

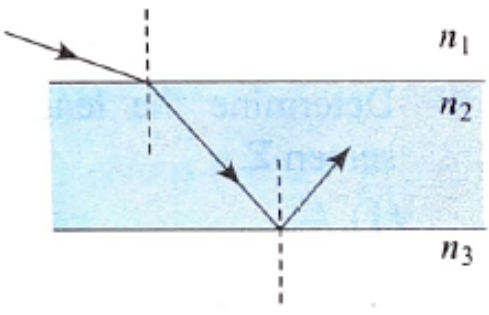
(B) $2L$

(C) $3L/2$

(D) $L/2$

Q.7

Light travelling through three transparent substances follows the path shown in figure. Arrange the indices of refraction in order from smallest to largest. Note that total internal reflection does occur on the bottom surface of medium 2.



(A) $n_1 < n_2 < n_3$

(B) $n_2 < n_1 < n_3$

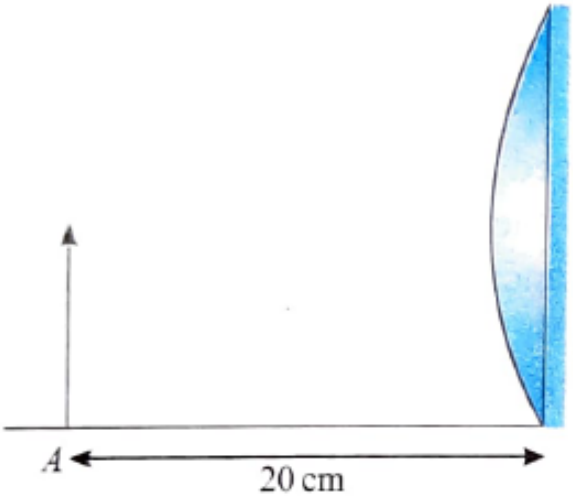
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(C) $n_1 < n_3 < n_2$

(D) $n_3 < n_1 < n_2$

Q.8

Focal length of the plano-convex lens is $15\ \text{cm}$. A small object is placed at A as shown in figure. The plane surface is silvered. The image will form at



(A) $60\ \text{cm}$ to the left of lens

(B) $12\ \text{cm}$ to the left of lens

(C) $60\ \text{cm}$ to the right of lens

(D) $30\ \text{cm}$ to the left of lens

Q.9

Let the $x - y$ plane be the boundary between two transparent media. Medium 1 in $z \geq 0$ has refractive index of $\sqrt{2}$ and medium 2 with $z < 0$ has a refractive index of $\sqrt{3}$. A ray of light in medium 1 given by the vector $\vec{A} = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}$ in incident on the plane of separation. The angle of refraction in medium 2 is

(A) 30°

(B) 45°

(C) 60°

(D) 75°

Q.10

The lateral magnification of the lens with an object located at two different positions u_1 and u_2 are m_1 and m_2, respectively. Then the focal length of the lens is

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(A) $f = \sqrt{m_1 m_2}\,(u_2 - u_1)$

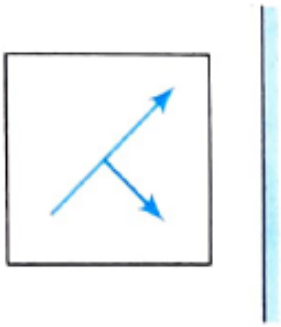
(B) $f = \sqrt{m_1 m_2}\,(u_2 + u_1)$

(C) $\frac{(u_2 - u_1)}{\sqrt{m_2 m_1}}$

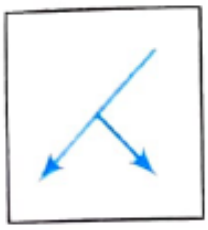
(D) $\frac{(u_2 m_2 - u_1 m_1)}{(m_2 - m_1)}$

Q.11

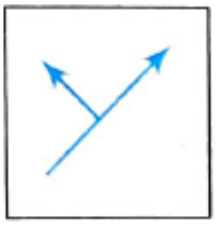
Choose the correct mirror image of object as shown in the following figure.



(A)



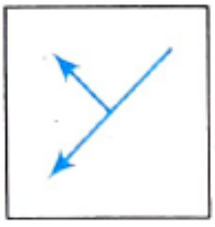
(B)



(C)



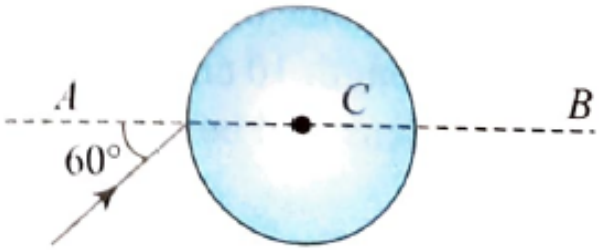
(D)



Q.12

A ray of light falls on a transparent sphere with centre at C as shown in figure,

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The ray emerges from the sphere parallel to line AB. The refractive index of the sphere is

(A) $\sqrt{2}$

(B) $\sqrt{3}$

(C) $3/2$

(D) $1/2$

Q.13

The critical angle for light going from medium X into medium Y is θ. The speed of light in medium X is v. The speed of light in medium Y is

(A) $v \cos\theta$

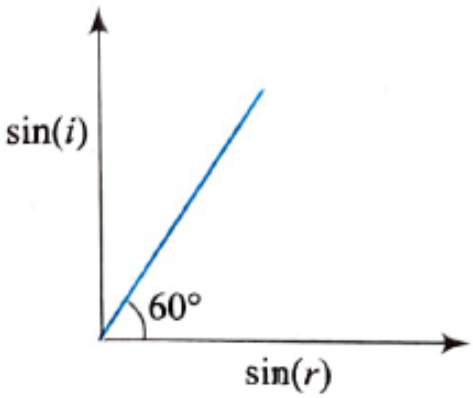
(B) $v/\cos\theta$

(C) $v \sin\theta$

(D) $v/\sin\theta$

Q.14

A ray of light is incident on a medium with angle of incidence i and refracted into a second medium with angle of refraction r. The graph of $\sin(i)$ vs. $\sin(r)$ is as shown in figure. Then, the velocity of light in the first medium is n times the velocity of light in the second. Find the value of $'n'$.



(A) $\sqrt{3}$

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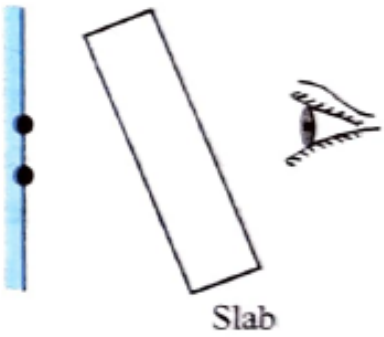
(B) $1/\sqrt{3}$

(C) $\sqrt{3}/2$

(D) $2/\sqrt{3}$

Q.15

The observer at O views two closely spaced spots on a vertical wall through an angled glass slab as shown in figure. As seen by observer, the spots appear



(A) shifted upward, such that distance between them remains same

(B) shifted downward, such that distance between them remains same

(C) spaced farther apart

(D) spaced closer together

Q.16

A ray of light travelling in the direction $\frac{1}{2}(\hat{i}+\sqrt{3}\hat{j})$ is incident on a plane mirror. After reflection, it travels along the direction $\frac{1}{2}(\hat{i}-\sqrt{3}\hat{j})$. The angle of incidence is

(A) 30°

(B) 45°

(C) 60°

(D) 75°

Q.17

Two identical glass $(\mu_x = 3/2)$ equiconvex lenses of focal length f are kept in contact. The space between the two lenses is filled with water $(\mu_n = 4/3)$. The focal length of the combination is

(A) f

(B) $\frac{f}{2}$

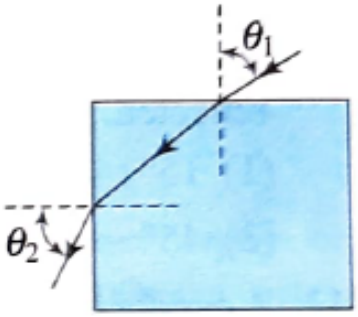
(C) $\frac{4f}{3}$

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(D) $\frac{3f}{4}$

Q.18

Light is incident on a glass block as shown in figure. If θ_1 is increased slightly, what happens to θ_2 ?



(A) θ_2 also increases slightly

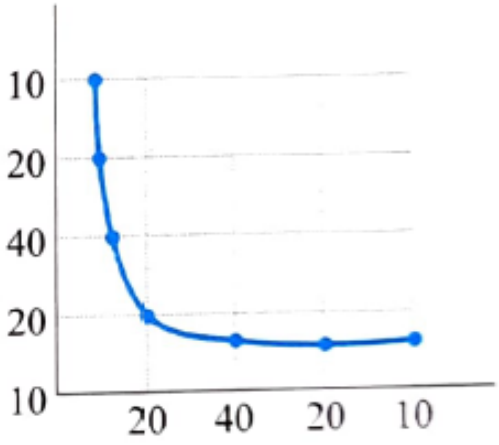
(B) θ_2 is unchanged

(C) θ_2 decreases slightly

(D) θ_2 changes abruptly, since the ray experiences total internal reflection

Q.19

A lens forms a real image of an object. The distance from the object to the lens is x cm and that from the lens to the image is y cm. The graph (see figure) shows the variation of y with x.



It can be deduced that the lens is

(A) converging and of focal length 10 cm

(B) converging and of focal length 20 cm

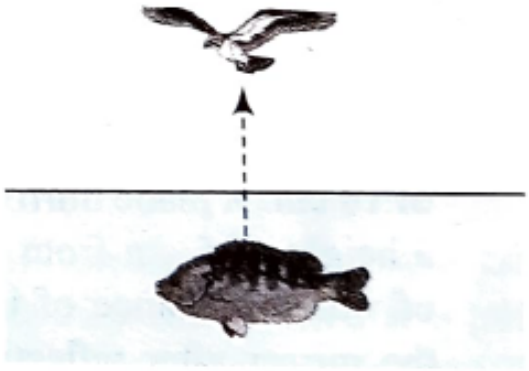
(C) converging and of focal length 40 cm

(D) diverging and of focal length 20 cm

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Q.20

A fish is vertically below a flying bird moving vertically down toward water surface. The bird will appear to the fish to be



(A) moving faster than its speed and also away from the real distance

(B) moving faster than its real speed and nearer than its real distance

(C) moving slower than its real speed and also nearer than its real distance

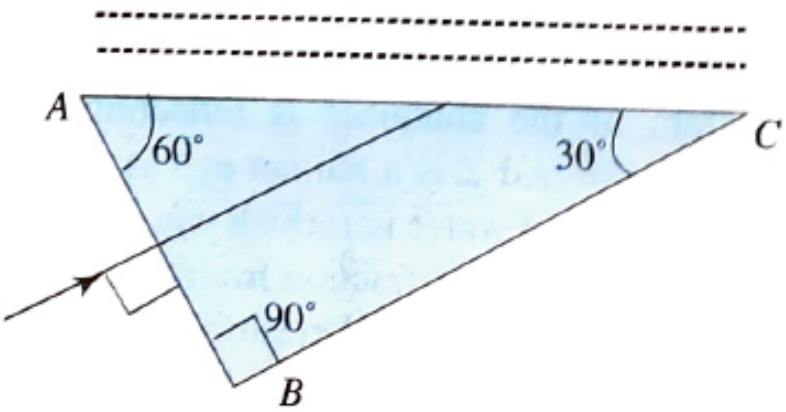
(D) moving slower than its real speed and away from the real distance

Q.21

How many of the following pairs are correctly matched?
I. Power of lens: diopter
II. Two waves of same frequency: diffraction and amplitude having constant phase difference
III. Total internal reflection: optical fibre
IV. Change of focal length of a lens with colour: spherical aberration

Q.22

As shown in figure, light is incident normally on one face of the prism. A liquid of refractive index μ is placed on the horizontal face AC. The refractive index of prism is 3/2. If total internal reflection taken place on face AC, m should be less than $\frac{I\sqrt{3}}{4}$, where I is an integer. Find the value of I.



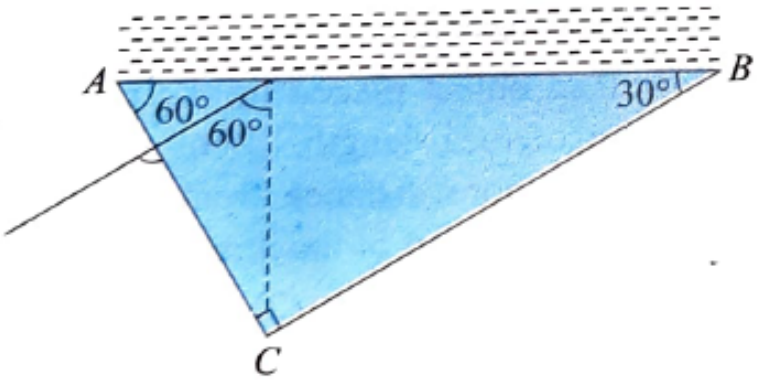
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Q.23

A circular beam of light of diameter $d = 2\ \text{cm}$ falls on a plane surface of glass. The angle of incidence is 60° and refractive index of glass is $\mu = 3/2$. The diameter of the refracted beam is _____? (Report the nearest integer)

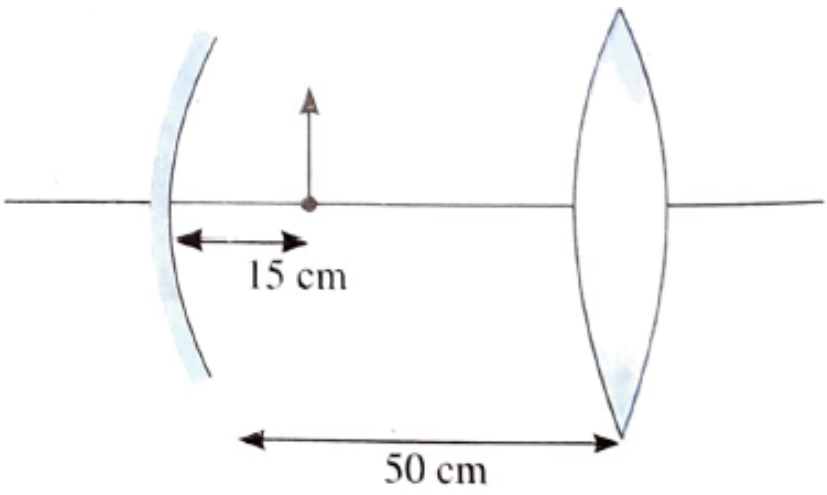
Q.24

ACB is right-angled prism with other angles as 60° and 30°. Refractive index of the prism is 1.5. AB has thin layer of liquid on it as shown. Light falls normally on the face AC. For total internal reflections, maximum refractive index of the liquid is (Round off the result to the nearest integer.)



Q.25

Consider a concave mirror and a convex lens (refractive index $= 1.5$) of focal length $10\ \text{cm}$ each, separated by 2 distance of $50\ \text{cm}$ in air (refractive index $= 1$) as showin the figure. An object is placed at a distance of $15\ \text{cm}$ from the mirror. Its erect image formed by this combination tas magnification M_1. When the set-up is kept in a medium of refractive index $7/6$, the magnification becomes M_2. The magnitude $\left|\frac{M_2}{M_1}\right|$ is _____



Q.26

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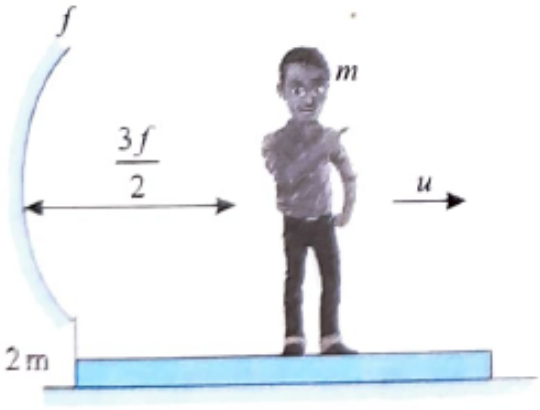
A candle is placed $20\ \text{cm}$ from the surface of a convex mirror and a plane mirror is also placed so that the virtual images in the two mirrors coincide. If the plane mirror is $12\ \text{cm}$ away from the object, what is the focal length of the convex mirror? (in cm)

Q.27

Diameter of plano-convex lens is $6\ \text{cm}$ and thickness at the centre is $3\ \text{mm}$. If speed of light in material of lens is $2 \times 10^8\ \text{m/s}$, the focal length of the lens is (in cm)

Q.28

A man of mass m starts moving w.r.t. a platform of mass $2\ \text{m}$ with a velocity $\mu = 9/13\ \text{m/s}$ as shown in the figure. The platform is fitted with a concave mirror of focal length f. The velocity of image $(m\ \text{m/s})$ at the initial moment is



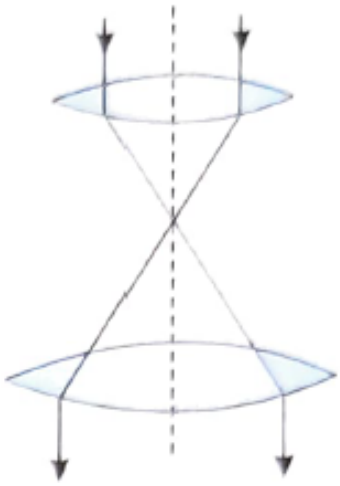
Q.29

The focal length of a thin biconvex lens is $20\ \text{cm}$. When an, object is moved from a distance of $25\ \text{cm}$ in front of it to 50 cm, the magnification of its image changes from $\text{m}_{25}\text{t}\ \text{m}50$. The ratio m_{25}/m_{50} is

Q.30

Consider a 'beam expander' which consists of two converging lenses of focal lengths $40\ \text{cm}$ and $100\ \text{cm}$ having a common optical axis. A laser beam of diameter $2\ \text{mm}$ is incident on the $40\ \text{cm}$ focal length lens. Then what is the diameter (in mm) of the final beam (see figure)?

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Answers & Solutions

Q.1 Answer:

$18 \text{ cm}, -27 \text{ cm}$

Solution:

$\frac{f_1}{f_2} = -\frac{\omega_1}{\omega_2} = -\frac{2}{3}$

Also $\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{F}$

$\Rightarrow \quad 1 + \frac{f_1}{f_2} = \frac{f_1}{F}$

$\Rightarrow \quad 1 - \frac{2}{3} = \frac{f_1}{54} \Rightarrow f_1 = 18 \text{ cm}$

$\therefore f_2 = -\frac{3f_1}{2} = -\frac{3\times 18}{2} = -27 \text{ cm}$

So the correct choice is (c).

Q.2 Answer:

The final image is formed at 60 cm from the lens towards right of it.

Solution:

$\text{u} = -15$
$\text{f} = 30 \text{ cm}$

from lens formula,

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$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} + \frac{1}{15} = \frac{1}{30}$

$\frac{1}{v} = \frac{-1}{30}$

$v = -30 \text{ cm}$

So distance from plane mirror is

$u_2 = 30 + 15 = 45 \text{ cm}$

So $u_2 = v_2$ for plane mirror

So from the lens $45 + 15 = 60$ cm to the right of it.

Q.3 Answer:

Myopia, concave, 50 cm

Solution:

Range of vision for healthy eye is 25 cm (near point) to ∞ (far point). If the person can see clearly only upto a maximum distance of 50 cm he is suffering from myopia (short sightedness). A shortsighted eye can see only nearer objects. This defect can be removed by using a concave lens of suitable focal length f = 50 cm.

Q.4 Answer:

only one image will be formed by the lens

Solution:

When a lens is painted as given, only the intensity of the image reduces since the portion blackened does not refract the incident light to form image.

Whereas the transparent still refracts as in case of a biconvex lens to obtain a single image following,

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Q.5 Answer:

$0.25 \text{ m}, 1 \text{ m}, 0.5 \text{ m}$

Solution:

Using the mirror equation, $\frac{1}{f} = \frac{1}{u} + \frac{v}{f}$,

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For point A, $\frac{1}{\mathrm{v_A}} = -1 + \frac{1}{3} = -\frac{2}{3}$

So, $\mathrm{v_A} = -\frac{3}{2} = -1.5$

Similarly for point B,

$\frac{1}{\mathrm{v_B}} = -1 + \frac{1}{5} = -\frac{4}{5}$

So, $v_\mathrm{A} = -\frac{5}{4} = -1.25$

the distance between two points is then $1.5 - 1.25 = 0.25$ m

Now, using the magnification relation,

$$\frac{\text{Image height}}{\text{Object height}} = -\frac{\text{Image distance}}{\text{Object distance}}$$

For point A, Imageheight $= -2 \times \frac{-1.5}{-3} = -1$ m

And for point B, Imageheight $= -2 \times \frac{-1.25}{-5} = -0.5$ m

Q.6 Answer:

$3L/2$

Solution:

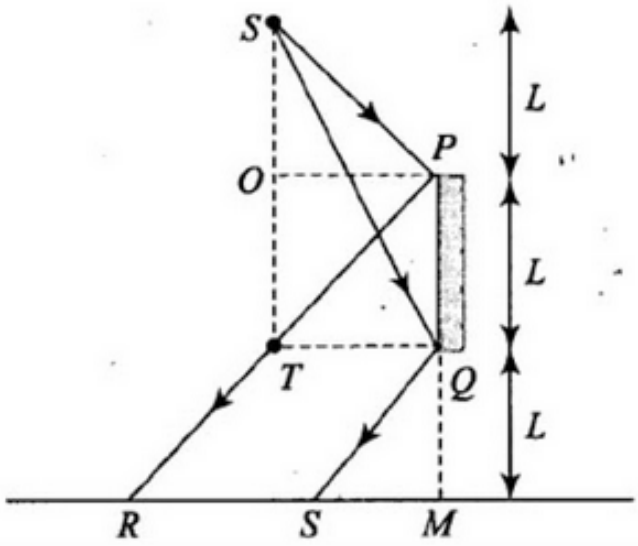
Ray diagram is as shown in the figure below:

Here, $\frac{\mathrm{SO}}{\mathrm{OP}} = \frac{\mathrm{PM}}{\mathrm{RM}} \Rightarrow \mathrm{RM} = 2\ \mathrm{L}$

$\mathrm{AlsO}, \frac{\mathrm{ST}}{\mathrm{TQ}} = \frac{\mathrm{QM}}{\mathrm{SM}} \Rightarrow \mathrm{SM} = \frac{\mathrm{L}}{2}$

$\mathrm{RS} = \mathrm{RM} - \mathrm{SM} = \frac{3\ \mathrm{L}}{2}$

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Q.7 Answer:

$n_3 < n_1 < n_2$

Solution:

As the ray moves toward the normal while entering medium 2 from 1 , we have $\mathrm{n_2} > \mathrm{n_1}$

For total internal reflection at interface of 2 and $3, \mathrm{n_2} > \mathrm{n_3}$**.**

Besides $\mathrm{n_3}$ **should also be less than** $\mathrm{n_1}$ **or else ray would have emerged in medium 3, parallel to its path in medium** 1**.**

Hence, $\mathrm{n_3} < \mathrm{n_1} < \mathrm{n_2}$ **is the correct order.**

Q.8 Answer:

12 cm to the left of lens

Solution:

A ray from point O **at first gets refracted from the curved surface of focal length,** $\mathrm{f_1} = 15$ cm **and then reflects from the plane mirror. Hence the plano- convex lens will become a curved mirror of focal length** F**.**

$\frac{1}{\mathrm{F}} = \frac{1}{\mathrm{f_m}} - \frac{2}{\mathrm{f_l}}$ **where** $\mathrm{f_m}$ **is focal length of mirror used.**

$\frac{1}{\mathrm{F}} = \frac{1}{\infty} - \frac{2}{15}$

$\mathrm{F} = \frac{-15}{2}$ cm

Using mirror formula, $\frac{1}{\mathrm{v}} + \frac{1}{\mathrm{u}} = \frac{1}{\mathrm{F}}$

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For $u = -20, \quad \frac{1}{v} + \frac{1}{-20} = \frac{-2}{15}$

Thus $\mathrm{v} = -12 \text{ cm}$

Thus the final image will form at 12 cm **to the left of lens.**

Q.9 Answer:

45°

Solution:

As the refractive index for $z > 0$ **and** $z \leq 0$ **is different,** $x - y$ **plane should be the boundary between the two media.**

The incident angle is the angle the vector makes with the zaxis (that is the direction of the normal to the x **y plane).**

Angle of incidence,

$$\cos i = \left| \frac{A_x}{\sqrt{A_i^2 + A_2^2 + A_\hbar^2}} \right|$$

$$\implies \mathrm{sos}\, 1 = \left| \frac{-10}{\sqrt{(6\sqrt{3})^2 + (8\sqrt{3})^2 + (10)^2}} \right|$$

$$\Rightarrow \cos \mathrm{i} = \left| \frac{-10}{\sqrt{108} + 192 + 100} \right|$$

$$\Rightarrow \cos \mathrm{i} = \left| \frac{-10}{\sqrt{400}} \right|$$

$$\implies \cos \mathrm{i} = \frac{1}{2}$$

$$\implies \mathrm{i} = 60$$

Now will apply the snell's law; we get

$\frac{\sin \mathrm{i}}{\sin \mathrm{r}} = \frac{\mu_2}{\mu_1}$**, where** μ_1, μ_2 **are the refractive index of medium 1 and 2 respectively and whose values are given in the question.**

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$$\implies \sin r = \frac{\sin i \times \mu_1}{\mu_2}$$
$$\Rightarrow \sin r = \frac{\sin(60) \times \sqrt{2}}{\sqrt{3}}$$
$$\Rightarrow \sin r = \frac{\frac{\sqrt{3}}{2} \times \sqrt{2}}{\sqrt{3}}$$
$$\implies \sin r = \frac{\sqrt{2}}{2}$$

multiply the numerator and denominator by $\sqrt{2}$, we get

$$\sin r = \frac{1}{\sqrt{2}}$$
$$\implies r = 45$$

is the angle of refraction in medium 2.

Q.10 Answer:

$\frac{(u_2 m_2 - u_1 m_1)}{(m_2 - m_1)}$

Solution:

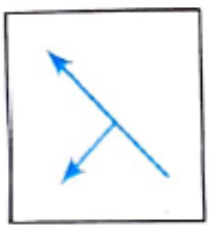
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$
$$u = -u; f = f$$
$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$
$$v = \frac{fu}{u-f}$$

Magnification is: $\frac{f}{u-f}$

$$\frac{m_1}{m_2} = \frac{\frac{f}{u_1 - f}}{\frac{f}{u_2 - f}}$$

$$f = \frac{u_2\, m_2 - u_1\, m_1}{m_2 - m_1}$$

Q.11 Answer:



Solution:

The right side of an object appears on the left side and the left side appears on the right side in case the mirror is placed in front of the object. In case the mirror is below the object or above the object, then the reflection or the image is obtained by turning the object upside down i.e. top

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goes to bottom and vice versa. Thus, following the logic the mirror image of the given figure is as shown in (C)

Q.12 Answer:

$\sqrt{3}$

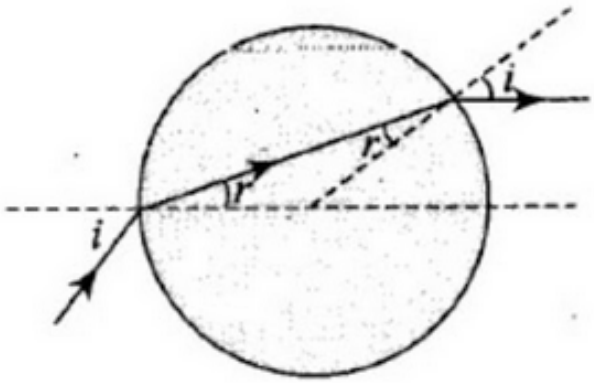
Solution:

Deviation by a sphere is $2(\mathrm{i}-)$

Here, deviation $\delta = \mathbf{60^\circ} = \mathbf{2(i - r)}$

or $\mathbf{i - r = 30^\circ}$

$$\therefore \mathrm{r} = \mathrm{i} - 30^\circ = 60^\circ - 30^\circ = 30^\circ$$
$$\therefore \mu = \frac{\sin \mathrm{i}}{\sin \mathrm{r}} = \frac{\sin 60^\circ}{\sin 30^\circ} = \sqrt{3}$$



Q.13 Answer:

$v/\sin\theta$

Solution:

Critical angle is θ

Let the refracting index of medium $\mathbf{X}$ **is** $\mathbf{n_x}$ **and that of medium** Y **is** $\mathrm{n_y}$

so, using Snell's law of refraction $\quad \mathbf{n_x \sin\theta = n_y \times 1} \quad -(1)$

Also, $\mathbf{v \times n_x = v_2 \times n_y}$

where $\mathbf{v_2}$ **is the speed of light in medium** Y

From (1), $\mathbf{v \times n_x = v_2 \times n_x \sin\theta}$

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Thus $\mathbf{v_2 = \frac{v}{\sin\theta}}$

Q.14 Answer:

$\sqrt{3}$

Solution:

Using Snell's law, $\frac{\mu_2}{\mu_1} = \frac{\sin i}{\sin} = \tan 60^\circ = \sqrt{3}$

We know, $\frac{v_1}{v_2} = \frac{\mu_2}{\mu_1} = n = \sqrt{3}$

option A is correct

Q.15 Answer:

shifted upward, such that distance between them remains same

Solution:

the two dots are just shifted towards the slab due to shift observed, the distance between them doesn't change and due to the incline view of the slab they are shifted upwards.

option A is correct

Q.16 Answer:

30°

Solution:

After reflection, the only component of the direction of propagation of incident light that is perpendicular to the plane mirror changes its sign Here, as the y component changes after reflection, we can assume that the direction of normal of the plane mirror is along with $\hat{j}$**. Therefore angle of incidence is angle between the unit vectors** $\frac{1}{2}(\hat{i} + \sqrt{3}\hat{j})$ **and** $\hat{j}$

$$\Rightarrow \cos(i) = \frac{1}{2}(\hat{i} + \sqrt{3}\hat{j})\hat{j}$$
$$\Rightarrow i = \cos^{-1}\frac{\sqrt{3}}{2} = 30^\circ$$

Q.17 Answer:

$\frac{3f}{4}$

Solution:

Water in between the lens can be considered to be a concave lens of refractive index of water as shown in the figure. This suggests that the radius of curvature of all the lenses is same.

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Consider the convex lens-

$\frac{1}{\mathrm{f}}=\frac{(\mu_{\mathrm{g}}-1)2}{\mathrm{R}}=\frac{\left(\frac{3}{2}-1\right)2}{\mathrm{R}}$

hence $\mathrm{R}=\mathrm{f}$

For the concave lens,

$$\text{Power }=\frac{1}{\mathrm{f}_1}=\left(\mu_{\text{water}}-1\right)\left(\frac{-2}{\mathrm{R}}\right)=\frac{-2}{3\mathrm{f}}$$

Power of the lenses get added,

$$\frac{1}{\mathrm{f}_{\text{combination}}}=\frac{1}{\mathrm{f}}+\frac{-2}{3\mathrm{f}}+\frac{1}{\mathrm{f}}$$

Hence $\mathrm{f}_{\text{combination}}=\frac{3\mathrm{f}}{4}$

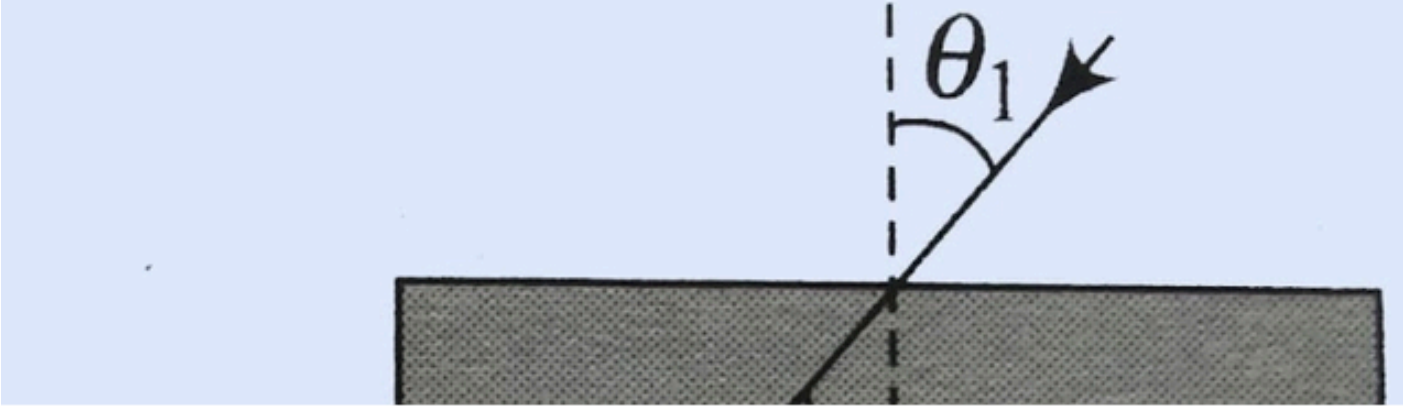
Q.18 Answer:

θ_2 decreases slightly

Solution:

As θ_1 **increases,** α **also inicreases. So,** $(90-\alpha)$ **decreases and hence** θ_2 **decreases.**

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Q.19 Answer:

converging and of focal length $10\ \text{cm}$

Solution:

Since the lens forms a real image, it is a converging lens.

Using the lens equation-

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

we get $\frac{1}{y} - \frac{1}{-x} = \frac{1}{f}$

From the plot we pick up any point to get the focal length of the lens.

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We pick up $x = 20, y = 20$

Hence we get $f = 10\ cm$

Q.20 Answer:

moving faster than its speed and also away from the real distance

Solution:

Refraction at a plane surface:

$$\frac{1}{\mu_{water}} = \frac{\text{real depth}}{\text{apparent depth}}$$

$$\text{similarly, } \frac{1}{\mu_{water}} = \frac{\text{real speed}}{\text{apparent speed}}$$

So,speed and distance both increases. therefore, bird will appear to be moving fast and also its distance will be more than the real distance.

Q.21 Answer:

—

Solution:

The power of lens, $P = \frac{1}{f}$**; where** f **is focal length and its unit is Dioptre.**

The diffraction of light occurs due to the superposition of secondary wavelets which are generated from various parts of a wavefront. Interference is the result of the superposition of light waves from two coherent sources.

Optical fibre works on the principle of total internal reflection since light rays fall inside the optical fibre at an angle greater than critical angle.

If R_1 **and** R_2 **are the radii of curvature of first and second refracting surfaces of a thin lens of focal length** f **and refractive index** μ

(refractive index of medium of lens w.r.t surrounding medium's refractive index), then using lens maker's formula,

$\frac{1}{f} = \mu - 1\frac{1}{R_1} - \frac{1}{R_2}$**.**

Here, $\mu = \frac{\mu_{glass}}{\mu_{surrounding\ \ medium}}$**.**

For different colours, μ **is different. Hence, lens is unable to bring all wavelengths to the same focal plane since wavelengths are focused at different positions in the focal plane.**

This phenomenon is known as chromatic aberration and not spherical aberration.

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Q.22 Answer:

—

Solution:

Critical angle between glass and liquid.

$\sin\theta_{\mathrm{C}} = \frac{\mu}{(3/2)} = \frac{2\mu}{3}$

Angle of incidence on $\mathrm{AC} = 60^\circ$

For TIR, i $> \theta_C$

$\Rightarrow \sin 60^\circ > \sin\theta_{\mathrm{c}} = \frac{2}{3}\mu$

$\mu < \frac{3\sqrt{3}}{4} = \frac{\mathrm{I}\sqrt{3}}{4}$ (given)

So, $\mathrm{I} = 3$.

Q.23 Answer:

—

Solution:

Let d' **be the diameter of refracted beam.**

Then,

$\mathrm{d} = \mathrm{PQ}\cos 60^\circ$

and $\mathrm{d}' = \mathrm{PQ}\cos \mathrm{r}$

i.e. $\frac{\mathrm{d}'}{\mathrm{d}} = \frac{\cos \mathrm{r}}{\cos 60^\circ} = 2\cos \mathrm{r}$

or $\mathrm{d}' = 2\,\mathrm{d}\cos \mathrm{r}$

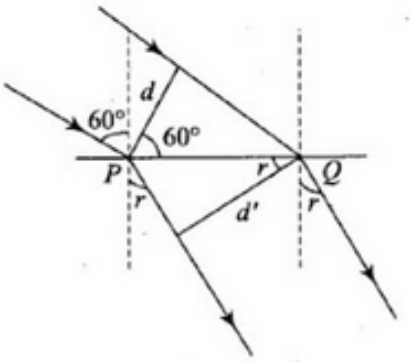
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$\sin r = \frac{\sin i}{\mu} = \frac{\sqrt{3}/2}{3/2} = \frac{1}{\sqrt{3}}$

$\therefore \cos r = \sqrt{1 - \sin^2 r} = \sqrt{\frac{2}{3}}$

$\therefore d' = (2)(2)\sqrt{\frac{2}{3}}$

$= 4\sqrt{\frac{2}{3}}$ cm ≈ 3.26 cm



Q.24 Answer:

—

Solution:

Clearly, $i_c \leq 60^\circ$

So, maximum possible value of $i_c = 60^\circ$

Now, $_1\mu_g = \frac{1}{\sin i_c}$ $\frac{\mu_g}{\mu_1} = \frac{1}{\sin i_c}$

or $\mu_1 = \mu_g \sin i_c = 1.5 \sin 60^\circ = 1.5 \times \frac{\sqrt{3}}{2}$

$= 1.5 \times 0.866 = 1.299 = 1.3$

Q.25 Answer:

—

Solution:

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The first image from reflection by spherical concave mirror can be found from mirror formula:

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f} \ldots$ (i)

For concave surface: $\mathrm{u}=-15 \mathrm{~cm}, \mathrm{f}=10 \mathrm{~cm}$ **Substituting in equation (i)**

$\frac{1}{v}+\frac{1}{-15}=\frac{1}{-10}=\frac{1}{v}=\frac{10-15}{150} \Rightarrow \mathrm{v}=-30 \mathrm{~cm}$

the magnification is given as $\mathrm{m}=-\frac{\mathrm{r}}{\mathrm{m}}=-\frac{-30}{-15}=-2$

The first image will act as object for lens and it's position can be found using lens formula

$\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \ldots$ (ii)

For concave surface: $\mathrm{u}=-20 \mathrm{~cm}, \mathrm{f}=10 \mathrm{~cm}$

Substituting in equation (ii)

$\frac{1}{v}-\frac{1}{-20}=\frac{1}{10} \Rightarrow \frac{1}{v} \frac{1}{10}-\frac{1}{20}=\frac{2-1}{20} \Rightarrow \mathrm{v}=20 \mathrm{~cm}$

the magnification is given as $\mathrm{m}=\frac{\mathrm{v}}{\mathrm{u}}=\frac{20}{-20}=-1$

total magnification $\mathrm{M}_1=-2 \times-1=2$

Now the whole set is immersed in liquid of $\mathrm{n}=\frac{7}{6}$ **the reflection from concave mirror will remain same as their is no change in object position and focal length.**

The change in focal length can be found using lens maker formula

$\frac{1}{f}=\left(\frac{\mu_z}{\mu_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

$\frac{1}{G_{ti}}=(1.5-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$**...(iiii)**

$\frac{1}{f_{\text{quil}}}=\left(\frac{1.5}{T}-1\right)\left(\frac{1}{R_4}-\frac{1}{R_2}\right) \ldots$ (**iv**)

dividing (iii) by (iv)

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$\frac{B_{pa1}}{t} = \frac{\frac{1}{2}}{\|f-1}$

The first image will act as object for lens and it's position can be found using lens formula

$\frac{1}{x} - \frac{1}{u} = \frac{1}{f} - \cdot(i)$

For concave surface: $\mathrm{n} = -20\ \mathrm{cm}, \mathrm{f} = \frac{\pi}{4}\ \mathrm{cm}$

Substituting in equation (ij)

$\frac{1}{\pi} - \frac{1}{-2!} = \frac{1}{\pi} \Rightarrow \frac{1}{2}\frac{1}{50} - \frac{1}{3} = \frac{9-70}{140} \Rightarrow \mathrm{v} = 140\ \mathrm{cm}$

total magnification $\mathrm{M}_2 = -2\mathrm{x} - 7 = 14$

$\frac{M_1}{M_1} = 7$

Q.26 Answer:

—

Solution:

as the image formed by convex mirror and by the plane mirror coincide the image distance formed by the convex mirror is 4 because, distance of image from plane mirror is 12 and the distance between plane mirror and convex mirror is 8 hence $12 - 8 = 4$

let f **be the focal lenght of the mirror,**

$\mathrm{u} = -20; \mathrm{v} = +4$

$\frac{1}{\mathrm{v}} + \frac{1}{\mathrm{u}} = \frac{1}{\mathrm{f}}$

$\frac{1}{4} - \frac{1}{20} = \frac{1}{\mathrm{f}}$

$\mathrm{f} = 5$

Q.27 Answer:

—

Solution:

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The lens is as shown in the figure.

To determine the radius of curvature R of the curved surface, we can write

$$\mathbf{R^2 = 3^2 + \left(R - \frac{3}{10}\right)^2}$$

Solving this, we get

$$\mathbf{R = 15.15}\mathrm{cm}$$

Speed of light in any medium is $\frac{\mathrm{c}}{\mu}$ where $\mathrm{c} = 3 \times 10^8\ \mathrm{m/s}$ and μ is the refractive index.

For the lens, $\frac{\mathrm{c}}{\mu} = 2 \times 10^8\ \mathrm{m/s}$

$\mu = \frac{3 \times 10^8}{2 \times 10^8} = 1.5$

Putting these values in the lens makers formula for plano convex lens, we have,

$\frac{1}{\mathrm{f}} = (\mu - 1)\left(\frac{1}{\mathrm{R}}\right) = (1.5 - 1)\left(\frac{1}{15.15}\right)\ \mathrm{cm}^{-1}$

Solving this, we get $\mathrm{f} \approx 30\ \mathrm{cm}$

Q.28 Answer:

—

Solution:

Using the relation $\mathrm{m} = \frac{\mathrm{f}}{\mathrm{f} - \mathrm{u}}$ we get

Magnification $\mathrm{m} = \frac{-\mathrm{f}}{-\mathrm{f} + \frac{3\mathrm{f}}{2}} \Rightarrow \mathrm{m} = -2 \Longrightarrow$

Taking the direction in the right hand side $\mathrm{V}_{\text{image/mirror}} = -\mathrm{m}^2\ \mathrm{V}_{\text{object /minrog}}$

$\mathrm{V}_{\text{image/mirror}}$ is velocity of image w.r.t. mirror

$\mathrm{V}_{\text{object/mirror}}$ is velocity of object wr.t mirror

$\Longrightarrow \mathrm{V}_{\text{image/mirror}} = -4\mathrm{u}$

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assume velocity of platform is v towards left

as the man is moving with velocity u towards right w.r.t. platform

$v_{\text{mas}} = v_{\text{man/plaform}} + v_{\text{platform}}$

$\mathrm{v_{trBa}} = \mathrm{u} - \mathrm{v}$

writing momentum balance equation

$\mathrm{m(u-v)} - 2\mathrm{mv} = 0 \Rightarrow \mathrm{v} = \mathrm{u}/3$

the velocity of platform = velocity of mirror

$\mathrm{V_{image/mirror}} = \mathrm{V_{image}} - \mathrm{V_{mirror}}$

$\mathrm{V_{image}}$ **is velocity of image w.r.t ground**

$\Longrightarrow \mathrm{V_{image}} = \mathrm{V_{lmuge\ /mim0t}} + \mathrm{V_{mirror}} = -4\mathrm{u} - \frac{\mathrm{u}}{3} = -\frac{13}{3}\mathrm{u}$

$V_{\text{lange}} = 3\ \mathrm{m/s}$

Q.29 Answer:

—

Solution:

given information $\mathrm{f} = 20\ \mathrm{cm}; \mathrm{u_{25}} = 25\ \mathrm{cm}; \mathrm{u_{50}} = 50\ \mathrm{cm}$

Formulas used $\frac{1}{\mathrm{v}} - \frac{1}{\mathrm{u}} = \frac{1}{\mathrm{f}}; \mathrm{m} = \frac{-\mathrm{v}}{\mathrm{u}}$

when $\mathrm{u} = -25\ \mathrm{cm}.$

$\Rightarrow \frac{1}{\mathrm{v}} = \frac{1}{\mathrm{f}} + \frac{1}{\mathrm{u}} = \frac{1}{20} + \frac{1}{-25} = \frac{1}{100} \Rightarrow \mathrm{V} = 100\ \mathrm{cm}$

$\mathrm{m_{25}} = \frac{-\mathrm{v}}{\mathrm{u}} = \frac{-100}{-25} = 4$

when $\mathrm{u} = -50\ \mathrm{cm}$

$\Rightarrow \frac{1}{\mathrm{v}} = \frac{1}{\mathrm{f}} + \frac{1}{\mathrm{u}} = \frac{1}{20} + \frac{1}{-50} = \frac{3}{100} \Rightarrow \mathrm{v} = \frac{100}{3}\ \mathrm{cm}$

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$m_{50} = -\frac{v}{u} = -\frac{1000}{3} \times -\frac{1}{50} = \frac{2}{3}$

Ratio of $m_{25} : m_{50} = \frac{4}{2/3} = 4 \times \frac{3}{2} = 6$

Q.30 Answer:

—

Solution:

Given,

Dia of beam, $2x = 2 \text{ mm}$

Focal length of lenses, $f_1 = 40 \text{ cm}$ **and** $f_2 = 100 \text{ cm}$

$$\frac{x}{f_1} = \frac{y}{f_2}$$
$$\Rightarrow \frac{1}{40} = \frac{y}{100}$$
$$\Rightarrow y = 5 \text{ mm}$$
$$2y = 10 \text{ mm}$$

Hence, Dia of final beam is 10 mm

y = 5mm

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